Riley Beenders

PRODUCTION RELIABILITY | INFRASTRUCTURE AUTOMATION | LINUX & WINDOWS SYSTEMS

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PROFESSIONAL SUMMARY

Engineer who builds the systems a production floor runs on, from Linux servers and container platforms to the networks and device fleets around them, and keeps them running once they are live. Automates the repetitive work, instruments what matters, and explains each failure and its fix plainly to vendors, operators, and leadership.

CORE EXPERTISE

Reliability & Operations: Production go-live and stabilization; root cause analysis; monitoring and alerting; troubleshooting across application, network, and hardware layers; third-tier escalation point; incident reports for leadership; operational documentation and user guides; vendor coordination; risk analysis and mitigation

Infrastructure & Networking: Linux (Debian) and Windows systems administration; Kubernetes; Docker and containerized applications; Proxmox virtualization; virtual machine and container management; reverse proxy and firewall routing; DNS; DHCP and IP address management; VLAN segmentation; Ubiquiti UniFi; multi-site network management; Raspberry Pi device fleets

Automation & Delivery: Python (asyncio, Starlette); PowerShell and WinGet; Windows Task Scheduler; SSH fleet tooling; GitHub Actions; Git and GitHub; continuous deployment on Vercel; self-updating services; Playwright

Software & Data: TypeScript; React and Next.js; Electron and Node.js; SQLite; Modbus TCP; Server-Sent Events; Microsoft Dynamics 365 Business Central (SaaS on Azure); data logging and production reports; LabVIEW

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA — January 2025 - Present

Lead, Research & Development Engineer

Proteor acquired Filament Innovations in January 2025, and the CTO role below continued as Lead R&D Engineer for the same product line, now Proteor Print.

- Own ExtrusionStudio, the plant's production data service, live on all three filament extrusion lines since October 2026 with no unplanned outage during production; rebuilt it from a LabVIEW proof of concept into a Python service on a headless Debian server.
- Fixed a lagging, disconnecting packing-bench panel during the ExtrusionStudio rollout by testing one layer at a time: halving the animation frame rate, checking for thermal throttling, and swapping in a fresh Raspberry Pi and SD card changed nothing; a direct Ethernet run fixed the dropouts and the label-printer delay, and running the 1440p touch screen at 1080p removed the lag.
- Took the Coplay location's Microsoft Dynamics 365 Business Central, Microsoft's SaaS ERP hosted on Azure, from evaluation to live production in five weeks, migrating more than 2,000 items and 150+ products, reconciling inventory counts on the floor, and syncing SKUs with Proteor's global locations. The whole facility of 10 to 15 people works in it, and 10 to 30 more at Proteor locations worldwide reference it.
- Traced the discrepancies in the 2026 Business Central inventory count, which I led across every item, to production orders processed without verification that never took items out of inventory, and helped move the plant to a more warehouse-focused workflow.
- Managed and expanded the division's IT infrastructure after the acquisition, standardizing software and internal processes to align with Proteor's company-wide systems.
- Serve as the plant's technical subject-matter expert, from industrial electronics on the extrusion lines to low-voltage sensors, and the escalation point for customer issues that go beyond what support can resolve, across mechanical faults, electrical faults, and user error.
- Work directly with equipment and software vendors to resolve issues on the Siemens PLC-controlled extrusion lines and the Duet3D printer controls, regularly into nights and early mornings.
- Contributed the machine requirements and integration for ProteorPrintOS, built with an external software partner on SimplyPrint, giving users remote operation, live video, AI-assisted failure alerts, and material, waste, and outcome metrics.
- Developing an offline version of the ICARUS-Lite and ICARUS-Pro control screen with over-the-air automated updates, so customers can run without a Wi-Fi or Ethernet connection.
- Write incident reports for leadership, and present project status, constraints, and budgets at Proteor's monthly USA review with leadership from its global locations.

- Estimate and track the budget of every engineering project, build its BOMs, and maintain each product's unit cost in Business Central.
- Performed risk analysis and mitigations to EU and U.S. standards across ProteorPrint hardware and cloud products.
- Owned the ICARUS-Lite, a compact industrial printer for prosthetic and orthotic production, through commercialization against unit goals of 60 in 2025 and 100 in 2026.
- Act as Hiring Manager for engineering roles, conducting technical interviews and owning final hiring decisions.

Filament Innovations | Whitehall, PA — May 2020 - January 2025

Across these roles, wrote customer proposals, scopes of work, and estimates, and began tracking project budgets.

Chief Technology Officer — January 2023 - January 2025

- Designed and hosted a redundant Kubernetes cluster running a self-developed ERP/CRM that centralized orders, inventory, tasks, projects, and customer interactions, generating BOMs dynamically and linking design data directly to purchasing.
- Oversaw the product-development pipeline, internal IT infrastructure, and technology strategy while keeping hands-on responsibility for mechanical, electrical, firmware, and software systems, and gave the CEO and the company regular updates as designs and software iterated.
- Led the design and launch of Gen3 POSEIDON, the flagship's production redesign, combining a 1 m³ build volume, filament-and-pellet dual extrusion, and braked Z drives that prevent uncontrolled motion during power loss.

Lead Electro-Mechanical Engineer — June 2022 - January 2023

- Integrated a multi-board CAN-bus control architecture that simplified wiring and debugging on larger, modular machines.
- Removed a manufacturing bottleneck with independently testable electronics enclosures that installed in minutes, coordinating custom harnesses, cut panels, and hardware improvements with vendors.

Industrial Engineer — April 2021 - June 2022

- Delivered THE KRAKEN, a 1 x 2 x 1.5 m hybrid pellet-and-filament gantry machine for a client producing the world's largest 3D-printed statue, carrying it from CAD, BOM, and sourcing through fabrication, assembly, and firmware tuning to customer support and operator training.

Engineering Technician — May 2020 - April 2021

- Diagnosed mechanical and electrical problems during assembly and operation, improving early product reliability.
- Tuned, calibrated, and optimized machine parameters while documenting design limitations and improvement opportunities.

ADDITIONAL RELEVANT EXPERIENCE

Northampton Area School District - Computer / Network Technician | June 2016 - August 2019 | Built network infrastructure and deployed more than 750 wireless access points across six school buildings; accelerated deployment of more than 2,500 Chromebooks with an Arduino-based tool that automated setup-data entry, and coordinated final device distribution with school administrators.

Blue Mountain Ski Resort - Ski Lift Operator (seasonal) | November 2019 - January 2021 | Operated lifts and assisted thousands of guests with ticketing, weather, and slope information; performed motor and control safety checks and supported the maintenance team during lift breakdowns.

Carver House Homeowners Association - Treasurer | April 2026 - Present | Manage the banking and monthly insurance payments for an association of about 80 units, and help establish a community-focused member board in place of a property management company, remaining Treasurer through a mid-year restructure.

EDUCATION & CERTIFICATIONS

Penn State University | B.S., Electro-Mechanical Engineering — December 2024

Machine Learning Onramp, MathWorks	2026
MATLAB Onramp, MathWorks	2026
Simulink Onramp, MathWorks	2026
Simscape Onramp, MathWorks	2026
Lean Six Sigma Yellow Belt, Educate 360	2025
Lean Six Sigma White Belt, Educate 360	2025
Certificate in Engineering Design and Tools, Penn State University	2023
Mechanical Engineering Certificate, Lafayette College	2019

SELECTED PROJECTS

ExtrusionStudio | Plant measurement and production system, Proteor — September 2025 - Present

Python, asyncio, Modbus TCP, Starlette, SQLite, Server-Sent Events, Raspberry Pi, Debian

Problem: Each of three extrusion lines had a tri-axial laser gauge, and the only record of a spool's diameter was whatever an operator wrote down. Production was speeding up, and the plant needed every reading captured, every spool judged against its tolerance, and the result visible on the floor at once.

- Runs one asyncio poller and Modbus TCP session per line, reading every line concurrently at 20 Hz so a powered-down line never stalls a healthy one; the poll rate is per-line configuration with a floor, because the right value depends on the PLC's response time.
- Built for supportability: Server-Sent Events that reconnect by themselves, no internet dependency on the floor, guides served by the server itself, a hand-written Modbus simulator that shares no code with the client, and a server that updates itself daily from its checked-out git branch so a fix can be tried on the real system.
- Added a calibration phase that times the filament from the laser to the spool, a delay of one to two minutes through the accumulator, so each reading and alarm lands on the spool it is actually wound on instead of firing at the laser.
- Logs every spool to its own CSV and a SQLite run table with a serial, QR label, and HTML certificate; print-on-demand thermal labels replaced full-sheet printing, removing up to about 30% of label waste per run.
- Runs a fleet of Raspberry Pi kiosks (projector, a touch screen per line, packing bench, statistics TV) that check in with the server and are updated, configured, and switched on and off with the work hours from one settings panel.

Unified UniFi Network | Network infrastructure, Proteor plus two residential sites — March 2024 - Present

Ubiquiti UniFi cloud gateways, VLANs, DHCP, DNS, firewall routing, network monitoring

Problem: As production grew, the plant's consumer mesh network assigned the same IP address to machines that came online together and never reliably detected or resolved the conflicts.

- Replaced it with a centrally managed UniFi network, using VLANs to separate production equipment, work devices, and IoT devices, and took central control of IP address assignment, DNS, VLANs, and port forwarding.
- Reduced IP conflicts and network-related downtime, improved security through segmentation, and gave room to add equipment; monitoring tools now surface connectivity problems so they are resolved faster, and all three sites are configured and monitored remotely from one interface.

RileyBeenders.com | Portfolio site and its automation, rileybeenders.com/about-this-site — June 2026 - Present

Next.js, React, TypeScript, Vercel, GitHub Actions, Playwright, Claude Code

Problem: Content, layout, and styling started tangled in one file, so every visual change risked the content, and the site had to keep shipping from a solo workflow.

- Split one resume JSON into focused data files merged at build time, edited from a local schema-driven Studio that runs on localhost only so it can never be deployed by accident, and deployed continuously on Vercel from the first holding page.
- A GitHub Actions workflow runs an agent on every push to main that refreshes the site's statistics and timeline from git and commits the result; Playwright scripts print and verify tailored resume PDFs from the same data the site renders.

BulkSSH | Windows desktop app, github.com/RileyBeenders/BulkSSH — October 2026 - Present

TypeScript, Electron, Node.js, SSH

Problem: Testing ExtrusionStudio's Pi kiosks meant repeating the same commands on every Pi, one SSH window at a time, after every change to the image.

- Opens many devices in a grid of live terminals, mirrors keystrokes to every ticked device, or broadcasts one command and collects each device's output and exit code, so an image change is tried on every Pi in one pass.
- Destructive commands such as reboot or rm -r need confirming before they reach several devices, host keys are never accepted silently, and dropped connections reconnect by themselves; released as an installer and a portable executable.

RB's Auto App Updater | Windows utility — September 2026

PowerShell, WinGet, Windows Task Scheduler, GitHub, SHA-256 verification

Problem: Windows applications update through different methods, so keeping dozens of them current meant visiting websites, downloading installers, and running each one by hand.

- Updates every WinGet-managed application with one double-click and no UAC prompt after a one-time setup, running the upgrades through an elevated scheduled task and upgrading each package on its own so a single failure never stops the rest.
- Translates WinGet exit codes into plain-language statuses such as security blocked, restart needed, and application in use, writes detailed logs that make remote troubleshooting possible, and updates itself only after the new script's SHA-256 hash matches its version manifest.